Fair Water

Group 6

2024-03-21

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1. Introduction

In this report, we present the culmination of our collaborative efforts as a team, working alongside FairWater researchers from Northumbrian Water, to address the pressing issue of water consumption and efficiency. Our project, undertaken in partnership with Newcastle University, aligns with the overarching goals of FairWater: to explore innovative solutions for enhancing water and energy sustainability in domestic environments.

Throughout our journey, we have delved into the intricate dynamics of household water usage, employing a range of analytical techniques and methodologies to gain insights into consumption patterns, identify inefficiencies, and propose actionable recommendations for behavior change. Drawing on the expertise of our diverse team members, which encompasses a wide array of skills ranging from data analysis and visualization to predictive modeling, we have endeavored to develop comprehensive solutions that cater to the needs of diverse stakeholders.

The framework used to obtain the results in the report is the **C**ross Industry **S**tandard **P**rocess for **D**ata **M**ining also known as **CRISP-DM**, performing 2 cycles of the model to obtain the results that will later on follow.

2. Business Understanding

The first step in our analysis process, focuses on understanding and defining the objectives from a business perspective, acknowledging stakeholders, providing clear aims and any assumptions made for the results and setting out success criteria to evaluate against. A preliminary plan was designed in order to achieve the objectives in context of the **CRISP-DM** framework.

2.1 Objective

The objective of this report is to present our findings and recommendations resulting from our collaborative project with FairWater led by Northumbrian Water, aimed at addressing the crucial issue of water consumption and efficiency in domestic settings. Moreover, we aim to produce insightful results, understand patterns, identify any problems, and give some recommended future solutions.

2.2 Research Question

Having in mind the objective we present the **3** questions that this report aims to answer. The questions are stated below.

1. **How can we effectively utilize analytical techniques and visualization tools to analyze household water consumption data, with the goal of uncovering underlying patterns, gaining valuable insights, and validating assumptions regarding water usage behavior?**
2. **What methodologies and approaches can be employed to disaggregate water usage data obtained from smart meters, allowing for the precise identification and analysis of individual water activities within households?**

3. **How can historical water consumption data and appliance signatures be leveraged to develop accurate predictive models for forecasting future water usage within domestic settings, and what considerations must be taken into account to ensure the reliability and applicability of these predictions?**

2.3 Data Mining & Success Criteria

Utilize the data which can be reviewed as accurate as possible in order to obtain desired results which can then be made available to the general public or any organization that needs to access them maintaining accuracy and authenticity so as to act as an information vessel. The results should meet the objective stated above, presented in a way which is easily accessible to stakeholders and can be explained with minimal effort.

3. Understanding Data

Moving on to phase 2 of the **CRISP-DM** cycle, and based on our objectives mentioned above, we consider the data requirements, as well as the available data to meet our objective goals. These are equally important in order to meet the objectives and generate useful results. After the initial data collection of the data set, the business objectives need to be reviewed, so as to ensure that the goals set are realistic with the available data.

3.1 Data Collection

In this report, data gathered from **WEUSEDTO** dataset. The data was collected from a 1 person household in Naples were readings were collected from **7** fixtures. The **7** fixtures include,Washbasin,Bidet,Kitchen Faucet,Shower,Washing Machine,Dishwasher,Toilet. The monitoring period was between **March-November 2019** and **July-October 2020**. The datasets were available in CSV format were each fixture had it's own dataset, as well as the data for the whole household was also provided. We collected in total 8 CSV files.

The 7 fixtures datasets was collect by the two methods, SWMs have been installed and programmed to collect high-resolution data on washbasin, bidet, shower and kitchen faucet at recording intervals of 1 seconds under operating condition and 5 seconds in case of no-use, For flush toilet and dishwasher, due to positional constraints, it was impossible to install the SWM. So, to register water consumption of the flush toilet the user is asked to push a button on its smartphone. An open source app has been configured to send to EmonCMS flush toilet cistern capacity value and dishwasher water use per cycle. The app allows for defining buttons shortcut to invoke HTTP hyperlinks.

Further details regarding the datasets can be found below included in the **Exploring Data** section. It's good to mention here that the following data sets can be found in the "data" file on "FairWater" file in **Project Template**.

3.2 Exploring Data

Due to the collection methods are different, the 8 dataseet has differnet attributes and different time dimension

End points	*Measured Yes/No*	*Methods*
Washbasin	✓	SWM
Bidet	✓	SWM
Shower	✓	SWM
Kitchen faucet	✓	SWM
Flush toilet	x	App HTTP request shortcuts
Dishwasher	x	App HTTP request shortcuts
Washing machine	✓	SWMs

Figure 1: Data Collection Methods

Table 1: Attributes of 8 Datasets

Dateset	Attributes 1	Attributes 2	Attributes 3	Time Collection
Wholehousehold	unix(unix)	flow (Ml/s)		One minute
Washbasin Bidet KitchenFaucet Shower Washing Machine	Time(unix)	Flow (Ml/s)		One seconds
Dishwasher Toilet	Time(unix)	Flow (Ml/s)	EndTime(unix)	No values between usage time

There are three main problems of the data set.

- Time select the day of two monitoring periods
- March – November 2019.
- July - October 2020.

Table 2: Time span of data collection

Season	2019	2020
spring	3, 4, 5	
summer	6, 7, 8	7, 8
autumn	9, 10	9, 10

During this time span, we have 8 months of data that was collected in 2019, and 4 months of data that was collected in 2020, 3 months spring data, 5 months summer data and 4 months autumn data.

4.Data Preparation

4.1 Data Cleaning

- Missing data

Due to inaccurate sensor readings we have 1-2 seconds missing. Considering the description of the sensor detecting that"One second sampling time and five minutes when no samples detected for five seconds". If the interval between the upper and lower time points is greater than 1 second and less than 5 seconds, the missing seconds between the two data will be supplemented, and the water flow value of the missing seconds equals to the subsequent water flow value. Based on this idea, we performed data imputation on data, other than toilets and dishwashers.

- Calculation water-flow per minutes of Toilet and Dishwasher

Collecting the data for toilet and dishwasher was not from the sensor but from the application records, so the water flow of these two fixtures were needed to be calculate. Created the minutes for time span of these fixture duration time, and calculated a minutes water flow of these two fixtures.

4.2 Integrate Data

- Merge the dataset

In order to make data statistics more easily, we merged the dataset together. First, we aggregated the 7 data sets respectively with one minute as the dimension. Then merged these data frames together, grouped the total data set by day, month, hour, minute.

- Filtering items

As for the filtering part, we filtered out the two main date span.According to the data set described and the data records, we chose the period of two time span: 2019-03-01 - 2019-11-01 & 2020-07-01 - 2020-11-01.

During this time span, we have 8 months of data that were collected in 2019, and 4 months of data that were collected in 2020, 3 months spring data, 5 months summer data and 4 months autumn data. Due to serious detecting time issues there are two problems with the dataset. One problem is that, within the same minutes the fixture contains water flow data in the minute-dimensional, but the wholehouse flow has no data. The other, is that a bunch of rows of data records is zero. Based on these two problems, we created a column named other that calculates the water flow of the other 7 fixtures, and make filtering to wholehouse or other column either of which is not 0.

4.3 Format Data

- Create new dataset

After we got the minutes time dimension dataset we then grouped the data set by hours and days respectively and created - exported three dataset with different time dimension for easy access: water_usage_by_min, water_usage_by_hour, water_usage_by_day.

The three new Datasets

- water_usage_by_day
- water_usage_by_hour
- water_usage_by_min

these three datasets are the merged datasets for water usage for wholehouse and different features. These three data sets have three different time dimensions. water_usage_by_day is the water usage data per day, water_usage_by_hour is the water usage per hour, and water_usage_by_min is the water usage data per minute.

In addition to the time dimension column, the entire household water flow, and the 7 fixtures water flow, two additional columns were added.

- Other =sum (features wateruasge flow except wholehouse)
- check_sum = wholehouse - Other

check_sum supposed to be equal 0, but due to data collection and missing values it has records.

The water flow of the whole house and the average flow and error of the 7 fixtures are as follows:

Table 3: Percentage Error of whole house dataset and 7 fixtures

Data	Mean Consumption mm/s
Whole House	67056.03
7 fixtures	66026.1
Percentage Error	1.53%

5.Modelling

5.1 Exploratory Data Analysis and Graphical Summaries

5.1.1 Overall water use trends

To uncover underlying patterns, derive valuable insights, and validate assumptions about water usage behavior, we will perform Exploratory Data Analysis (EDA) on the dataset obtained through data cleaning.

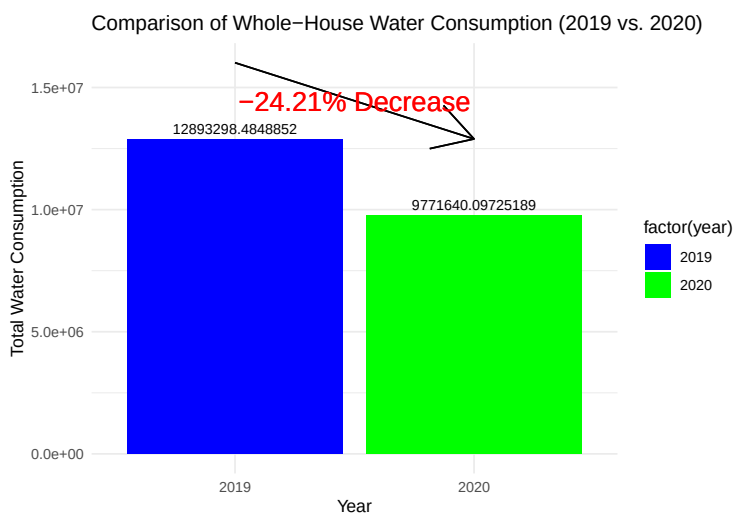

Figure 2: Bar plot showing the total water consumption between the 2 years.

Comparing Total Water Consumption between 2019 and 2020 Looking at **Figure** 2, we start our visualization with a very informative bar plot that indicates the difference in total water usage between the years 2019 and 2020. Our findings revealed that in 2019, the total water consumption across all monitored months was higher compared to 2020. Upon further examination, we calculated a percentage decrease of approximately **24.13%** in total water consumption from 2019 to 2020. While this decrease may initially appear significant, it is crucial to consider the subtle detail in our analysis.

One critical factor to note is the difference in monitoring periods between the two years. The monitoring period for 2020 was shorter, standing at 5 not monitored months, which could have influenced the observed

decrease in total water consumption. Considering this aspect, it is plausible that the apparent decrease in water usage from 2019 to 2020 may be partly because of the shorter monitoring duration in 2020. In our report we will be delivering a more in depth analysis on this aspect, however now we are stating the first few facts.

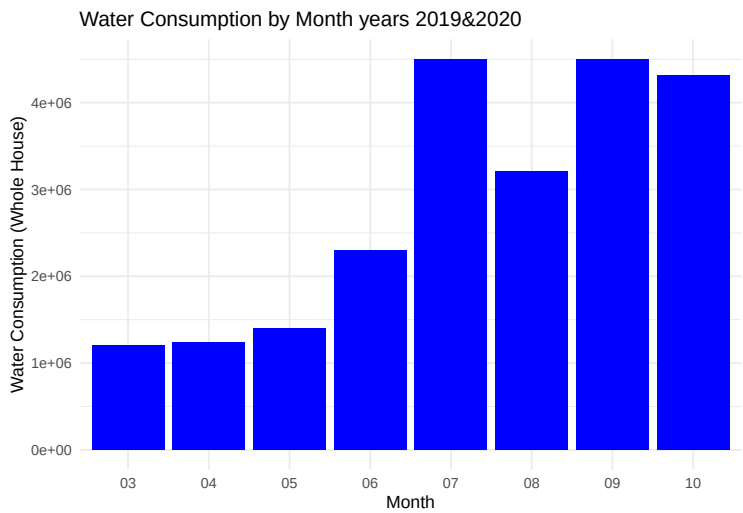

Figure 3: Bar plot showing total water consumption throughout the months for both years.

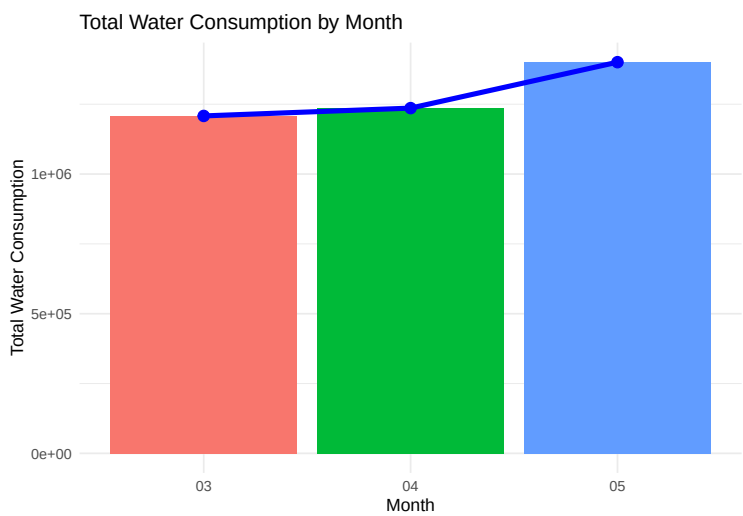

Figure 4: Bar plot indicating an increasing trend for water consumption for year 2019 during spring months

Total Water Consumption by Month Going a step further, in **Figure** 3 we are investigating if there is a relationship of total water consumed and month. Looking at figure 2 we are positive to say that there is a noticeable increase of water consumed as we head into the summer season that included months **6,7,8 and 9**. The large increase seen during these months is explained by the hotter weather during those months and without any further analysis in this section we assume that the increase was from the more frequent showers taken throughout the days. There is a small decrease during August (month 8) however this does not change the fact that water consumption increased during the summer months.

It is also good to note here that the large increase might also be explained from the additional readings from the year 2020 where the reporting of data begun in July (month 7). However, up until May (month 6) were

only the 2019 data was monitored during the first 4 months, we can clearly see a smooth increase of water consumption as we move on to the summer season. This is clearly indicated in **Figure** 4 , were an increasing trend is visible, and one can say that there is a positive correlation between total water consumption and the different season months.

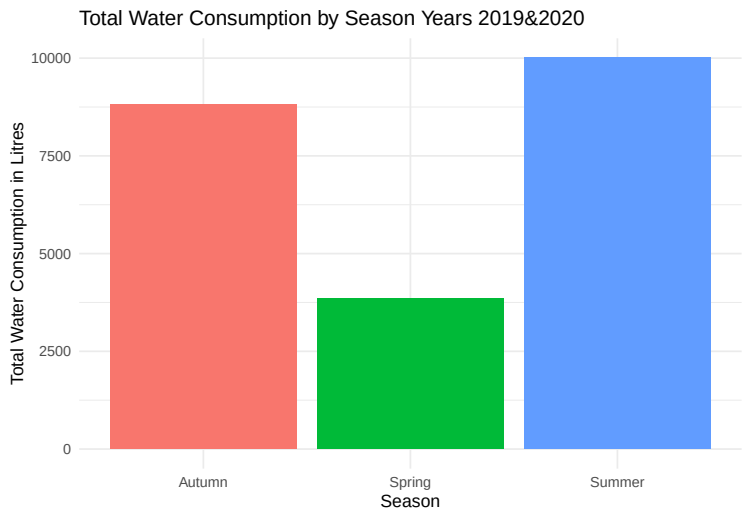

Figure 5: Total water consumption by Season for the 2 monitored years

Total Water Consumption by Season In figure 5, "Total Water Consumption by Season" reflects the quantity of water that was consumed through the following seasons, **Autumn, Spring, Summer**. Total water consumption is given on the vertical axis. The proportion of total water consumption by season is also given on the x- axis.

The bar chart indicates that water consumption is at it's peak during the **Summer season**, where the blue bar touches almost 10,000 Litres. Following second is Autumn, where the red bar is stands above 7,500 Litres. Spring shows the least water consumption, with the green bar at around 2,500 Litres. It is also good to note here that during the sprin season only data for the year 2019 was available hence the lower consumption. The water consumption pattern for this season could be also governed by the factors like the variation of temperature, different lengths of daylight, and different human activities or agricultural demands that are regulated through each type of season. The data presented is indicative of a marked rise in water use during the warmer months, likely due to greater irrigation and cooling.

Further Analysis The "Monthly Water Consumption Comparison" which is represented in figure 6, compares the water consumption of the two different years in the four common months: July (07), August (08), September (09), and October (10). On the y-axis: total water consumption, while the x-axis represents months. The bar representing the year 2019 is red in color, and that for the year 2020 is blue.

In this visualization, each pair of bars represents a month, with adjacent red and blue bars for comparison. In each of the months presented, it reveals that the consumption of water for the year 2020 was higher compared to that of 2019. It is pointing towards an annual rise from July to September, leveling off or gently dropping into October. The greatest difference between these two years seems to occur in the ninth month, where the blue bar for 2020 is way taller than the one for 2019. This comparison, in turn, will help in trend analysis of water usage, determination of likely causes for the increased consumption, or how effective year-on-year measures for water conservation have been.

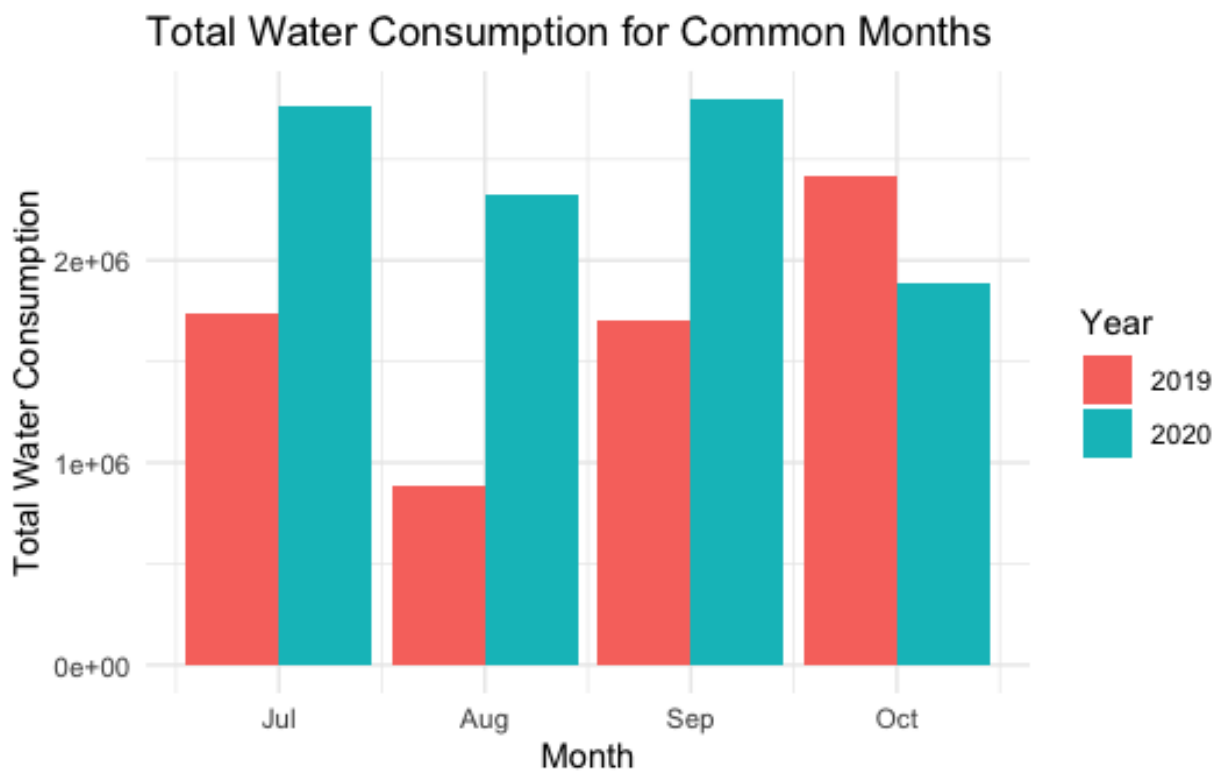

Figure 6: Bar plot showing water consumption during the common months that data was monitored

5.1.2 Fixtures water usage trends

7 Fixtures Water Exhausted Proportion of Total Waterusage To identify and analyze individual water activities within households, it's essential to grasp the proportion of water consumption attributed to each fixture in the overall usage. This understanding enables a deeper insight into the water consumption behaviors of household members.

From Figure 7, we observe the primary household water consumption fixtures, ranked by usage, as follows: showers account for 30%, toilets for 22%, washbasins for 14%, and kitchen faucets also for 14%. However, it's worth noting the specific recording method for toilet data. Moving forward, our focus will primarily be on investigating the usage habits of the remaining three fixtures.

4 main features water usage hourly total From Figure 8,the total water usage of kitchenfaucet has three peaks at 8 a.m., 12 a.m. and 8 p.m. corresponds to the user's breakfast lunch and dinner times. The total shower of usage has two peaks at 7 a.m. and during 3 p.m. to 5 p.m. which shows people usually shower during this time period. The total usage of toilet only have peak at 7 a.m. and other times of the year, while the frequency at other times appears to be relatively consistent or average.The total usage of Washbasin has a peak at 7 a.m., the other water consumption is not that different.

Shower Use Acitities First, focusing on shower data, from Figure 9, we observe that the average bathing duration ranges from 2 to 5 minutes. The preferred bathing times are typically between 6-7 a.m. and 3-5 p.m. Interestingly, shower usage at 6:00 and 7:00 a.m. is slightly lower compared to usage at 3:00, 4:00, and 5:00 p.m., suggesting shorter shower durations in the morning.

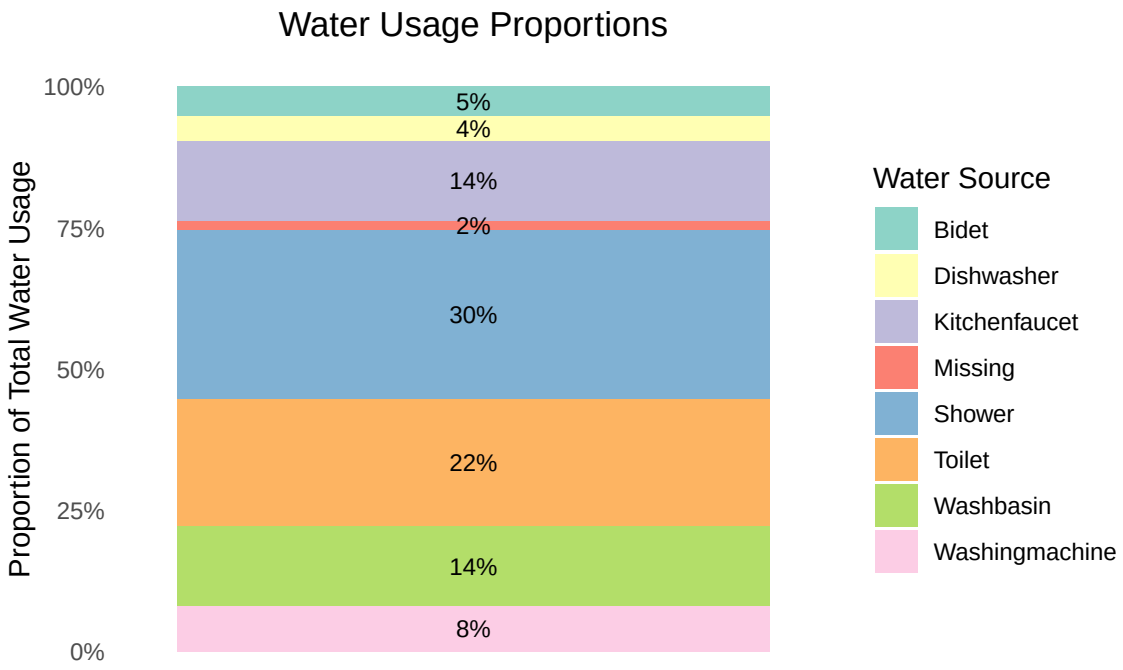

Figure 7: Proportion of Total Water Usage

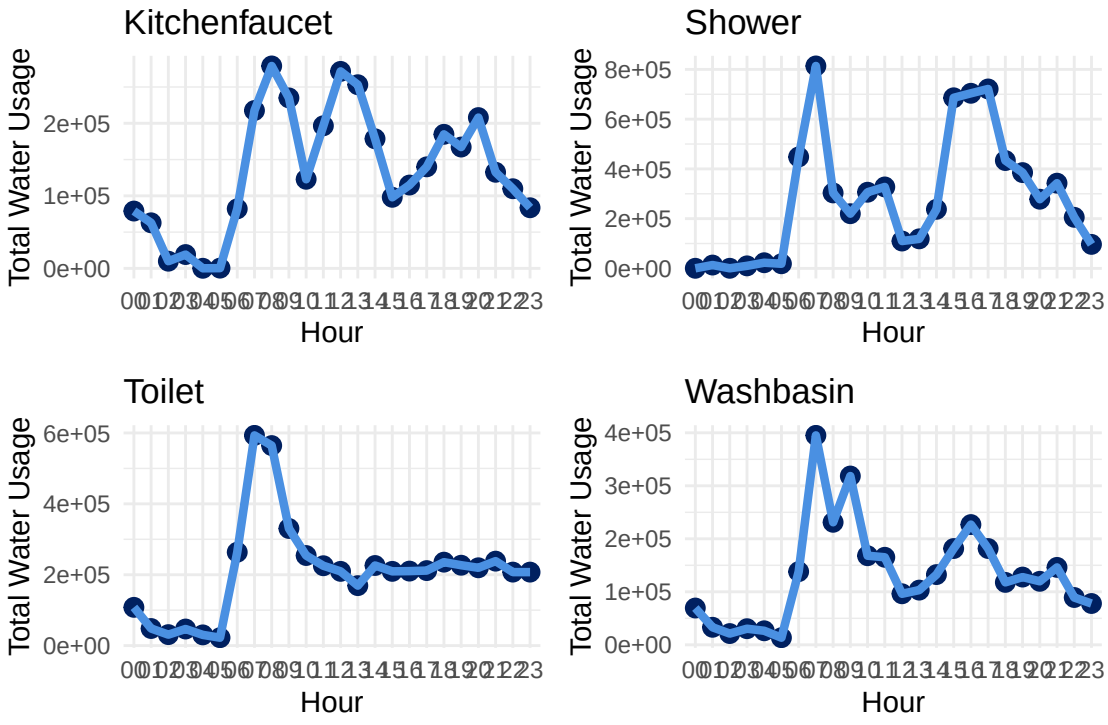

Figure 8: 4 main features water usage hourly totals

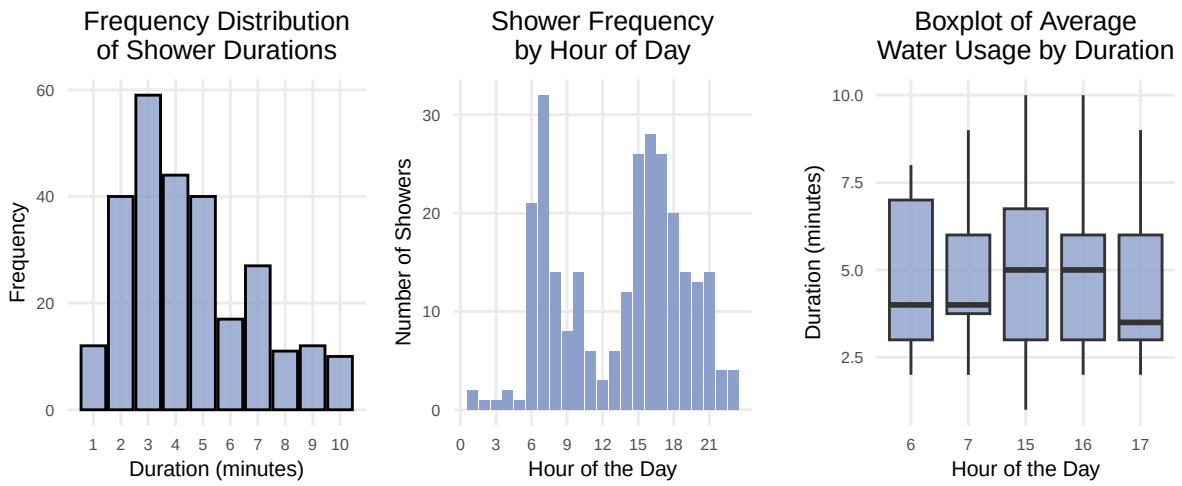

Figure 9: Frequency shower of Duration Mintues

Toilet Use Acitities Next, focusing on the toilet dataset, as shown in Figure 10, we observe a higher frequency of toilet usage between 7 and 8 o'clock, while the frequency at other times appears to be relatively consistent or average.

Regarding the washbasin data, as shown in Figure 11, individuals typically use the washbasin for approximately 1 minute. The peak usage occurs at **7 a.m.**, while usage throughout the rest of the day remains relatively consistent. Notably, when the usage duration ranges from 1 to 3 minutes, the average flow rate of the faucet is notably higher compared to other times. This phenomenon may be attributed to individuals habitually maximizing the faucet flow rate early on to expedite tasks, resulting in a higher average flow rate within a short time frame.

Kitchen Faucet Use Acitities Regarding the kitchen faucet data, as illustrated in Figure 12, individuals typically utilize the kitchen faucet having duration of approximately one minute. There are three peak usage periods: 7-8 a.m., 12-1 p.m., and 7-8 p.m. These coincide with meal preparation and post-meal cleanup times. Notably, when the usage duration ranges from 1 to 2 minutes, the average flow volume is higher, mirroring the pattern observed with the washbasin. This is likely due to individuals habitually maximizing faucet flow early on to expedite tasks, resulting in a higher average flow rate over a brief period.

The average time of use of the equipment and the frequency of use at each point in time, according to which water pressure can be appropriately lowered during these peak hours to reduce water consumption, and by analysing the data on time and frequency of use, it is possible to identify the time periods with the greatest potential for water saving and to develop water saving education and guidance campaigns targeting these time periods. For example, if it is found that bathing peaks in the morning, users can be encouraged to bathe during off-peak hours, and some form of incentive can be offered to encourage this behavioural change. The use of water-saving taps can significantly reduce water flow and can be effective in saving water even during peak usage hours. Consideration can also be given to installing water-saving taps, which often incorporate advanced water-saving technologies, such as air dosing, that can reduce water use while maintaining the pressure of the water flow. This can be done by setting timers or smart home system reminders to help users realise when they are using their taps. Timed reminders can be sent out 2-3 minutes before they are reached, encouraging users to complete their usage as soon as possible to reduce water consumption.

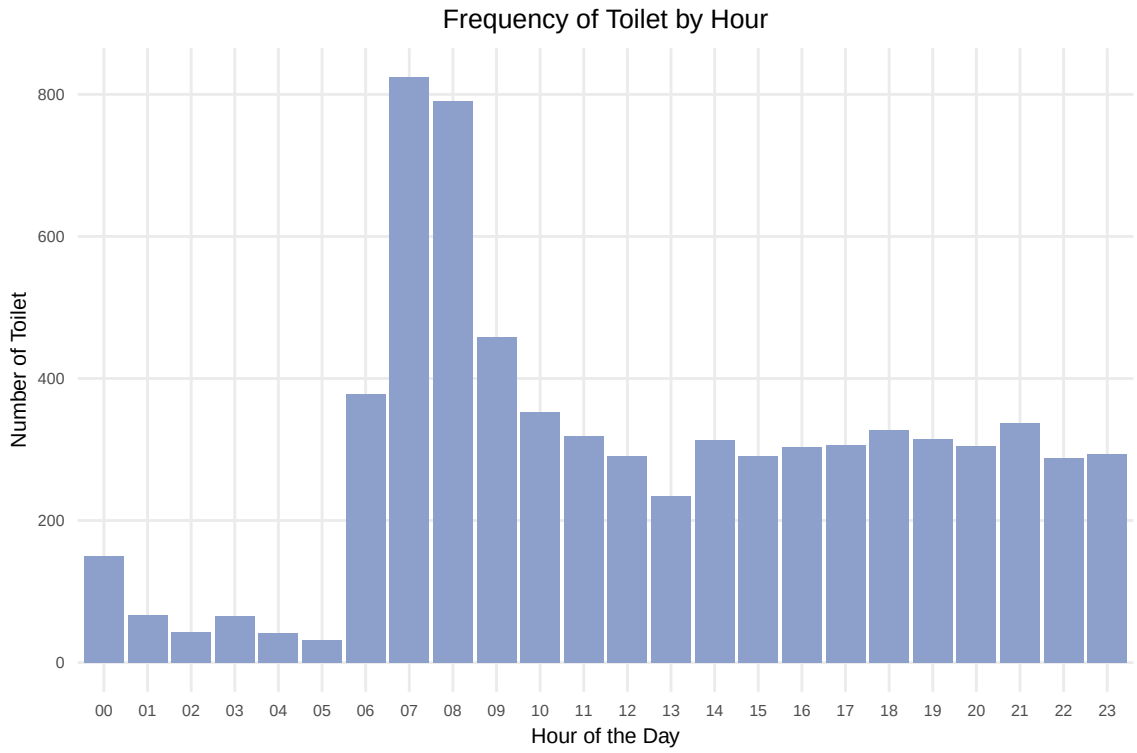

Figure 10: Frequency of Toilet by Hour

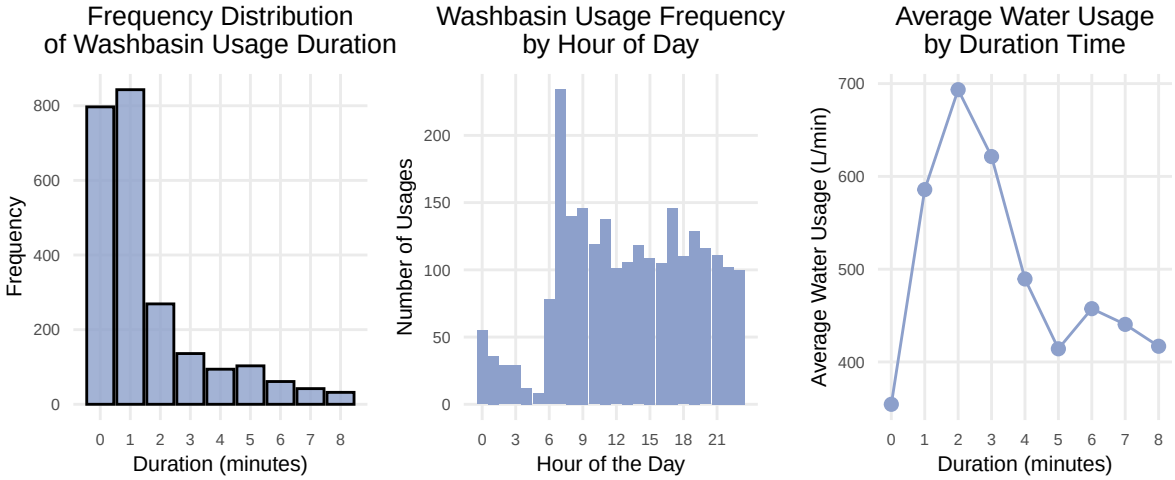

Figure 11: Frequency shower of Duration Mintues

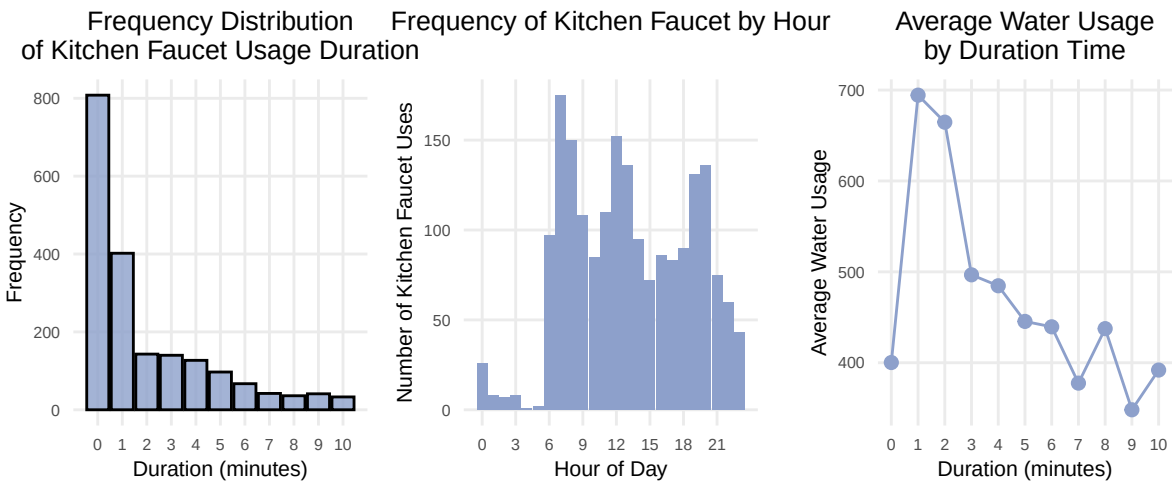

Figure 12: Frequency shower of Duration Mintues

5.2 Statistical Learing Method & Data Minning

5.2.1 Human hourly water usage activities on weekday and weekends & K-Means

The second aspect we aim to investigate is people's water usage behavior throughout the day, particularly examining differences between working days and weekends. As seen in Figure 13, we observe that peak water consumption on weekdays typically occurs around 7 o'clock. In contrast, the overall water consumption trend on weekends appears to be more subdued, with a lower peak observed at 8-9 o'clock.

The data indicates marked variances in human water usage patterns across different hours of the day, with further disparities evident between weekdays and weekends. This trend underscores the importance of considering the time factor as a crucial aspect in analyzing water consumption behaviors.

Next, the time periods were analysed using k-means clustering algorithm for clustering of the time periods, which further helped us to identify the water use patterns at different times of the day. And through bar charts, we can identify the percentage of these activities in each cluster to understand the primary and secondary water use activities. Finally, we preforming behavioural pattern recognition to analyse the specific causes of water spikes, such as whether they are related to household members' home time, eating habits or cleaning habits, which can reveal potential opportunities to improve water use efficiency.

Table 4: k-means of weeksday table result

cluster	Bidet	Dishwasher	Kitchenfaucet	Shower	Toilet	Washbasin	WashingmachineAvg_Hour	
1	259.21	483.10	1117.38	2095.55	1580.90	743.61	0.01	20.48
2	290.13	271.01	1202.54	2942.85	1311.94	1075.77	192.89	14.45
3	579.86	200.38	922.95	2664.33	2155.09	1378.41	1147.89	8.36
4	232.22	220.32	514.81	95.95	558.72	488.66	0.00	1.75

From Table 4 and Figure 14 , we can find that on weeksdays

- Peak water use in the evening (around 20:45): Water use behaviour during this time is characterized by higher water use activities in shower (2022.61) and kitchenfaucet (1028.76), which may be related to people's habits of cleaning and sleeping after dinner.
- Peak afternoon water use (around 14:30): The behaviour of water use in the afternoon shows showering, kitchen water use and more general water use activities. However, the unusually high water use for showering, much more than any other time period, and the relatively high washingmachine relative to the other time periods, may imply that there is a specific water use activity in the afternoon time, such as showering after a household member has returned from being away, or that intensive cleaning activities may have been carried out during this time period.

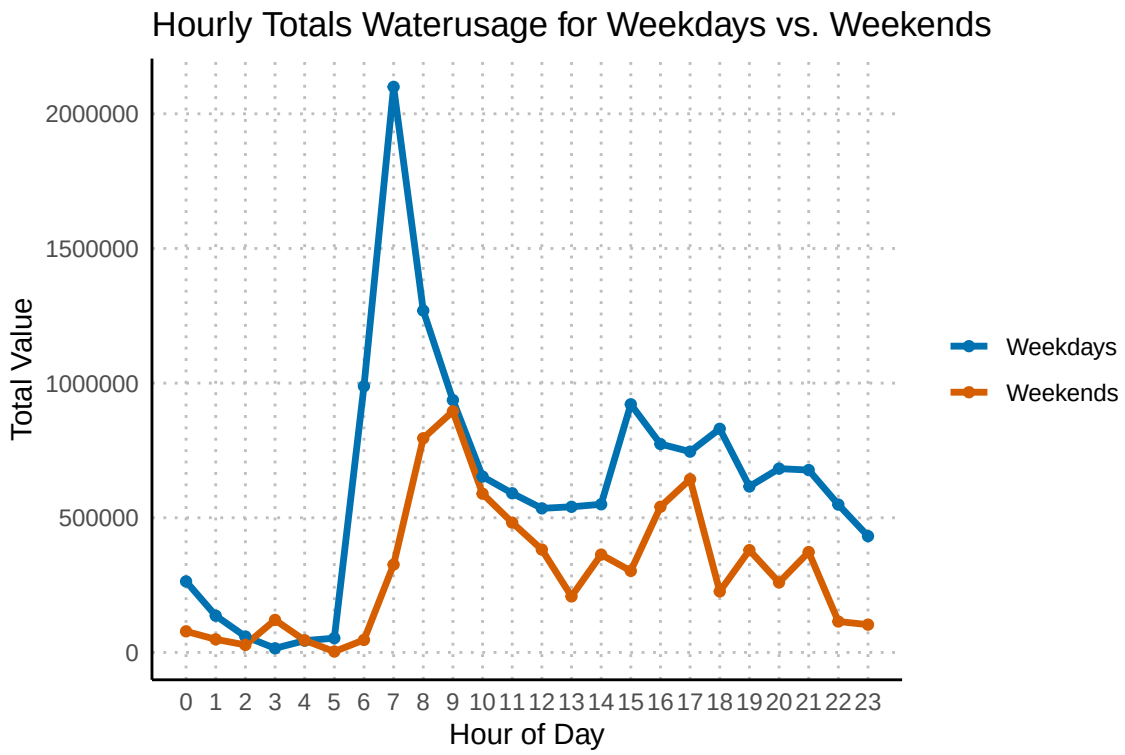

Figure 13: Hourly Totals Waterusage for Weekdays vs. Weekends

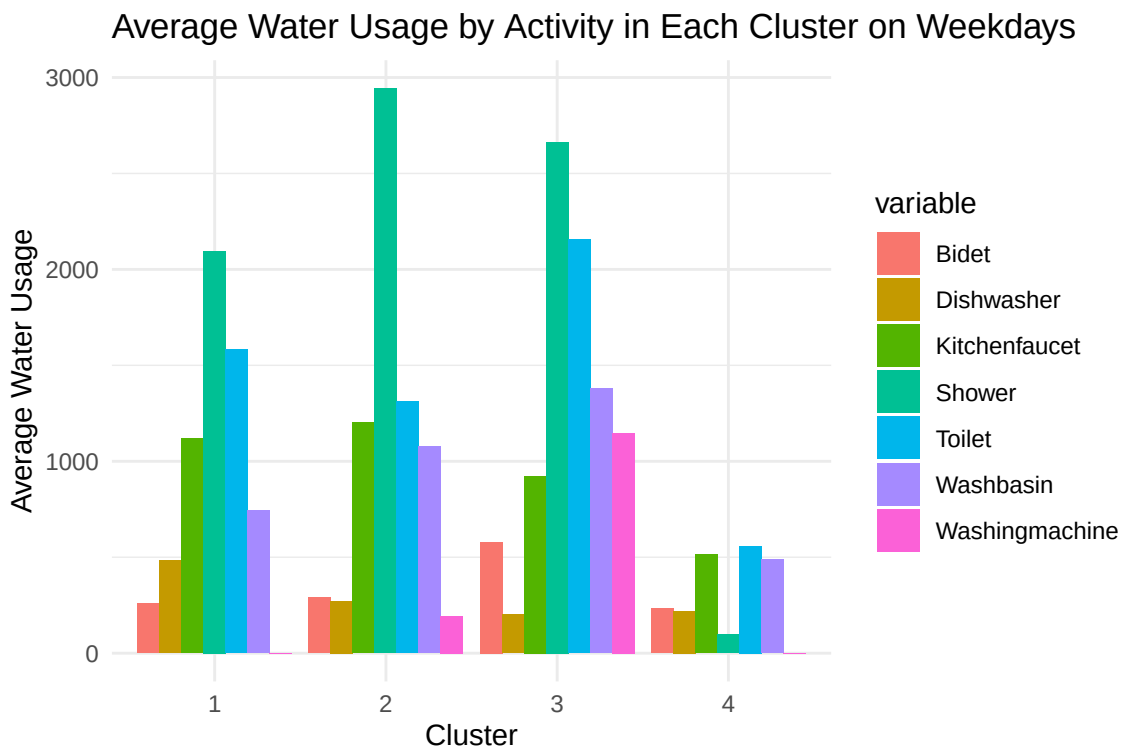

Figure 14: Average Water Usage by Activity in Each Cluster on Weekdays

Table 5: k-means result on Weekends

cluster	Bidet	Dishwasher	Kitchenfaucet	Shower	Toilet	Washbasin	Washingmachine	Avg_Hour
1	474.57	405.21	1612.83	1578.57	1989.04	1167.81	1998.52	10.28
2	358.40	123.28	355.90	169.80	1181.01	472.76	0.00	2.89
3	274.43	329.77	846.57	2978.75	1597.28	1116.90	0.13	18.74

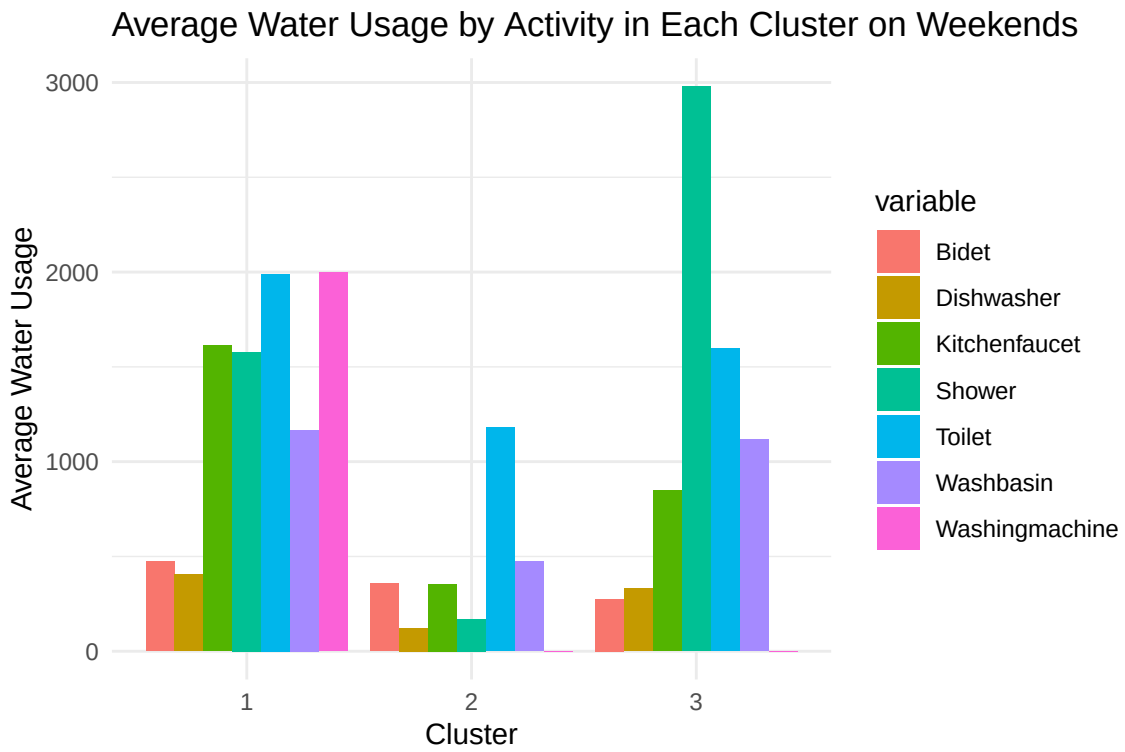

Figure 15: Average Water Usage by Activity in Each Cluster on Weekends

As for weekends, From Table 5 and Figure 15,

- Peak morning water use (around 10:18): This time period is notably characterised by high water use in Washingmachine, toilet, kitchenfaucet and shower. In addition, washing machine water use is much higher than other peak water use periods. This may be due to the fact that weekends are usually a time when many households do a lot of cleaning, including laundry and home hygiene cleaning, which can lead to increased water use for washing machines and toilets. And people may tend to get up a little later in the week and shower after breakfast, which can lead to increased water use at kitchen taps and showers.
- Peak afternoon water use (around 14:53): Water use is generally low at 3pm, particularly for washing machines, which use zero water, with toilets being the only water activity with a high level of water use. This suggests that household water use activities are infrequent at this time of day, and may be a time when family members are out of the house or taking a break. Also, some households may be water-conscious and choose to use their washing machines at times when water costs are lower (e.g. at night).
- Peak night water use (around 18:42): At this point in time, water use is highest in the SHOWER and WASHBASIN, probably due to the evening peak in bathing when families return home. toilets and washbasins also have higher water use, related to hygiene needs in the evening.

Based on the above analysis, here are some preliminary recommendations to improve water efficiency: 1. Time-differentiated water use incentives: Introducing a differential tariff system that makes it more expensive to use water at peak times and less expensive to use water at lower peak times, thus encouraging users to engage in high volume water use activities when water consumption is low. 2. Peak pricing: Make water use more costly during peak hours (afternoon) by implementing a policy of peak tariffs to incentivise residents to use water during lower-cost hours. Installation of water-saving devices: Install water-saving showerheads, faucets, and dual-flush toilets in homes for high water-using appliances, such as showers and toilets.

5.2.2 Fixture use correlation & Hierarchical Clustering & Association Rule Analysis

After examining the activities of individuals across different hours, we aim to explore any potential correlations between the usage of these fixtures. To address this question, we utilize hierarchical clustering to analyze 7 fixtures in the water_use_by_hour dataset

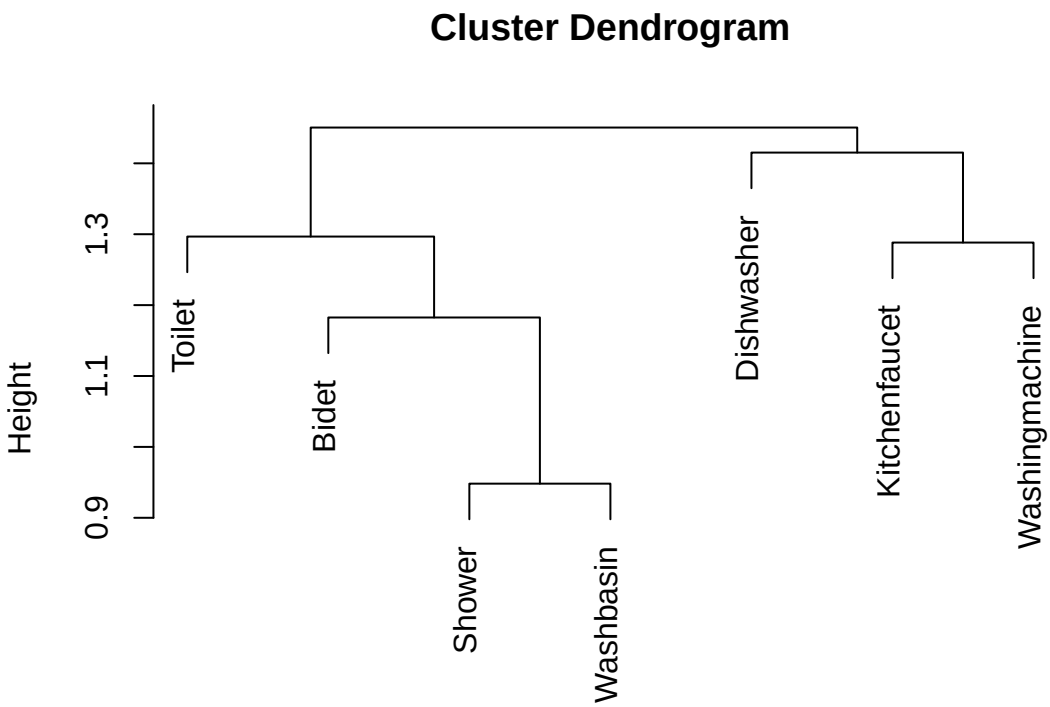

Figure 16: Cluster Dendrogram of 7 fixtures

Referring to Figure 16, hierarchical clustering has partitioned the 7 fixtures into two distinct clusters. One cluster comprises fixtures related to personal hygiene, including shower, washbasin, bidet, and toilet, while the other cluster likely pertains to water usage for housework activities, such as kitchen faucet, washing machine, and dishwasher.

Furthermore, the hierarchical structure suggests a closer relationship within the personal hygiene cluster compared to the housework activities cluster. This implies that people tend to use personal hygiene fixtures

concurrently or within a similar time frame.

To delve deeper into the usage patterns of these fixtures, we categorized each hour's water consumption as either personal hygiene or housework activity. Subsequently, we conducted association rule learning analyses separately for these two datasets to uncover potential usage correlations within each category.

Table 6: Associatio rule analysis result

	lhs	Var.2	rhs	support	confidence	coverage	lift	count
[1]	{Bidet, Shower}	=>	{Washbasin}	0.1151631	0.8633094	0.1333973	1.364361	360
[2]	{Bidet, Toilet}	=>	{Washbasin}	0.1250800	0.8408602	0.1487524	1.328882	391
[3]	{Shower}	=>	{Washbasin}	0.1801024	0.8019943	0.2245681	1.267459	563

According to Table 6, we observed three rules within the personal hygiene fixture category. These rules indicate an 86.12% confidence that when Bidet and Shower are used simultaneously, Washbasin will also be used. Similarly, when Bidet and Toilet are used simultaneously, there is an 84.08% confidence that Washbasin will be used. Additionally, when Shower is used alone, there is an 80.08% confidence that Washbasin will also be used. However, for housework activities, no significant usage connections were observed among them.

The high level of confidence in these usage patterns suggests that closer integration of these facilities may lead to water-efficient designs. For example, designing a bathroom layout where water from a hand basin can be used for a toilet or bidet may significantly reduce water use. Energy and water efficiency could be improved by developing predictive water heating systems that learn from these usage patterns to ensure that hot water is only prepared when it is likely to be needed, thereby reducing water and energy wastage.

6. 2nd CRISP-DM

6.1 Understanding Data

6.1.1 Describe data

```
Data Summary for aggregatedWholeHouse:
                unix           flow
count  1.660820e+05  166082.000000
mean   1.574215e+09     138.141072
std    1.859693e+07     656.585100
min    1.550048e+09       0.000000
25%    1.558866e+09       0.000000
50%    1.567153e+09       0.000000
75%    1.596442e+09       0.000000
max    1.604424e+09   20052.000000

Missing Values Check:
unix    0
flow    0
dtype: int64
```

Figure 17: Data Summary

Dataset Overview Our dataset encompasses a significant volume of 166,082 observations related to the "flow" variable, positioning us with a robust foundation for time series analysis. The "unix" field has been

adeptly transformed into datetime format and designated as the index, aligning with best practices for time series investigation.

Missing Values Assessment: The meticulous evaluation for missing values reveals an ideal scenario; neither the "unix" nor the "flow" fields exhibit any gaps. This absence of missing data exempts us from the need for preliminary pre-processing steps such as imputation or record elimination, allowing an immediate transition to the analytical phase.

Flow Data Distribution Insights: The descriptive analysis unveils a mean flow value of 138.14 juxtaposed with a median (50% quantile) of 0, indicating a potential high skewness in the data with predominant low flow occurrences interspersed by sporadic surges. A standard deviation of 656.59 underscores the extensive variability in flow readings. The stark disparity between the maximum flow value of 20,052 and the 75% quantile of 0 further accentuates the existence of exceptional peak events, hinting at specific incidents or anomalies warranting further examination.

Proposed Analytical Directions:

- Trend Analysis: Preliminary statistics have shed light on the dataset's characteristics, yet an exploration of temporal trends remains untapped. Visualizing the time series could unravel patterns of water utilization over time, such as identifying a progressive increase or decrease in consumption year on year.
- Seasonality and Periodicity Exploration: Given the dataset's comprehensive span, potentially covering several years, it warrants an investigation into seasonal behaviors (for instance, heightened water usage during certain months) or daily cyclical trends (like increased consumption at specific times).
- Outlier Detection Imperative: The detected peaks underscore the importance of identifying outliers, which could signify data recording inaccuracies, equipment malfunctions, or legitimate consumption spikes. Discerning the nature of these anomalies is paramount for a precise time series analysis.

6.1.2 Exploraing data

- Peaks and Fluctuations Analysis:

The visual representation of the flow data elucidates pronounced volatility, characterized by numerous significant peaks. This pattern suggests intermittent surges in water consumption, possibly linked to specific occasions or time frames.

- Cyclical Behavior Insight:

Amidst the notable fluctuations, a discernible cyclical pattern emerges, particularly in the latter segments of the graph, where certain regularities become apparent. This observation hints at a rhythm in water usage behavior, potentially reflecting weekly or monthly cycles.

- Outlier Investigation:

The data exhibits extreme peaks at certain junctures, especially noticeable towards the right of the chart. These values starkly exceed the general range of observations and necessitate a thorough examination to ascertain their nature—whether they constitute outliers, errors in data capture, or genuine instances of increased water demand.

- Time Series Stability Overview:

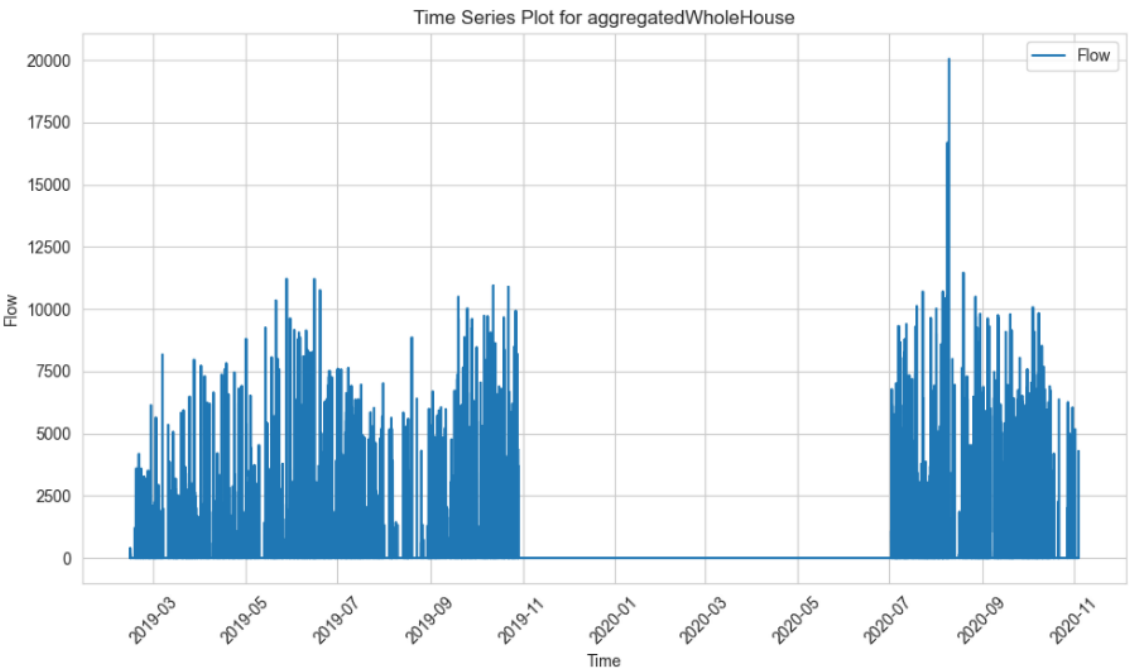

Figure 18: Wholehouse Flow Time Series

Overall, the graphical analysis portrays a relatively consistent flow range over time, with the exception of the aforementioned peaks. This observation indicates the absence of a persistent long-term trend in water consumption, barring the occurrence of sporadic spikes.

- Seasonal Influence Consideration:

Notable within several cycles are heightened flow levels, which may suggest the presence of seasonal influences, such as variations in temperature or holiday periods, affecting water usage.

- Dataset Continuity Confirmation:

The displayed graphs, spanning from March 2019 to November 2020, reveal no gaps in the timeline, affirming the dataset's continuity throughout the observed period. This continuity underlines the dataset's suitability for comprehensive time series analysis.

6.2 Data Preparation

Based on the insights gleaned from the time series analyses and the need to develop a robust model, we have segmented our dataset for model training and validation purposes. Specifically, we designated the data spanning from March to November 2019 as our training set. This segment will serve as the foundation for our model, enabling it to discern and assimilate potential patterns and cyclical behaviors in water usage. On the other hand, the data from July to November 2020 has been allocated as our validation set. This will facilitate the assessment of the model's performance across distinct time frames, affirm its predictive capabilities, and ascertain its generalizability. By adopting this approach, we aim to bolster our confidence in employing the model for future projections of water consumption.

6.2.1 Data Cleaning

The data pre-processing phase encompasses a series of critical steps:

- Reading the Data: We employ pd.read_csv to import a CSV file into a pandas DataFrame from the specified file path. By setting index_col='datetime', we designate the 'datetime' column as the DataFrame's index. Furthermore, the parse_dates=['datetime'] argument is pivotal for converting string representations in the 'datetime' column into DateTime objects suitable for pandas. This conversion is essential in time-series analysis, facilitating efficient temporal slicing and the extraction of temporal features.
- Eliminating Unnecessary Columns: We identified and removed the 'unix' column, a redundant timestamp feature, in favor of the 'datetime' column. Extracting Time Features: The add_time_features function is instrumental in deriving time-related attributes such as hour, day of the week, and month from the datetime index. These extracted features are invaluable for the model to discern seasonal and daily patterns.
- Compiling Feature List: A comprehensive list named features is compiled, encompassing all the feature names. This list ensures uniformity throughout the normalization process and model training phase.
- Data Normalization: We utilize the MinMaxScaler to normalize the data, scaling all numerical features to the range [0, 1]. Normalization is a crucial step, particularly for neural network models, as it facilitates faster convergence and enhances model performance.
- Constructing Input-Output Sequences: The create_sequences function is designed to generate input and output sequences. In time-series forecasting, a sequence of consecutive time points is typically required as input to predict the subsequent value(s). These sequences form the foundation for training the model.
- Setting Sequence Lengths and Feature Indices: The variables input_sequence_length and output_sequence_length specify the length of the historical data window observed by the model and the forecast horizon, respectively. The flow_feature_index and time_feature_indices pinpoint the specific features within the model's inputs that are predictive of flow.
- Reshaping Input Dimensions: To meet the model's input requirements, we adjust the input dimensions, expanding them as needed. Neural network architectures generally demand a three-dimensional input format, encapsulated as (number of samples, time steps, number of features).

6.3 Modeling

6.3.1 Mixed Neural Network Model Build

This architecture is a sophisticated hybrid neural network that synergistically combines the strengths of a Convolutional Neural Network (CNN) and a Long Short-Term Memory (LSTM) network. Below, I delve into the intricacies of its design and the advantages it offers:

- Processing of Flow Feature Input:

This segment is dedicated to processing a singular feature: the flow (denoted as flow_input). It employs an LSTM layer, renowned for its efficacy in time series forecasting due to its proficiency in capturing long-term dependencies within sequence data. The deployment of 128 cells within this layer ensures that each temporal step is represented in a 128-dimensional output, facilitating the retention and discernment of intricate patterns in the input sequence.

- Time Feature Input Processing Segment:

This section is tasked with handling additional temporal features such as hours and days of the week. The data initially traverses a Conv1D layer, commonly utilized for local feature extraction in sequential data.

Subsequently, it passes through a MaxPooling1D layer, which serves to diminish data dimensionality and parameter count while preserving pivotal features. A Flatten layer then transitions the 2D output into a 1D format, streamlining the data for subsequent processing layers.

- Integration of Flow and Temporal Features:

The fusion of the LSTM and Flatten layer outputs is achieved via concatenation, enabling the model to concurrently consider flow and temporal features in its predictions. This integration harnesses both the long-term dependencies inherent in traffic data and the local dependencies of other temporal features, enhancing the model's predictive capacity.

- Fully Connected Layer Implementation:

A Dense layer is employed here to amalgamate and decipher non-linear relationships among diverse features. This layer, containing 100 neurons and utilizing the ReLU activation function, introduces an added layer of nonlinearity to the model.

- Output Layer Functionality:

The terminal Dense layer is tasked with producing the final forecast, tailored to predict a series of 24 time points. This layer omits an activation function, aligning with the regression nature of the problem where the objective is to forecast continuous traffic flow values.

6.3.2 Mixed Neural Network Model Advantage

- Feature Learning Proficiency: The integration of CNNs with LSTMs empowers the model to not only capture extensive temporal dependencies but also autonomously abstract significant local patterns within the input features.
- Parameter Efficiency: The CNN layer's capacity to derive effective feature representations with minimal parameters, coupled with the pooling layer's role in reducing computational demands and mitigating overfitting, underscores the model's efficiency.
- Model Adaptability: The hybrid framework's flexibility allows for straightforward modifications to accommodate various time series challenges by adjusting input sequence lengths, feature dimensions, or network layer configurations.

In essence, this hybrid model excels in delivering robust performance and versatility when addressing time series datasets with multifaceted features and complex dynamics. Its capability to forecast future sequences across multiple time points within a given window renders it invaluable for multi-step forecasting tasks

6.4 Evaluation

6.4.1 Evaluate Results

Model Training Diagnostic Chart

- **Training Loss (train_loss):** This metric illustrates the training set's loss diminishing over successive training cycles, as depicted in the graph. The consistent decrease in this value signifies the model's effective learning from the training data, evidenced by the gradual reduction of errors.

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
time_input (InputLayer)	(None, 48, 5)	0	-
conv1d (Conv1D)	(None, 46, 64)	1,024	time_input[0][0]
flow_input (InputLayer)	(None, 48, 1)	0	-
max_pooling1d (MaxPooling1D)	(None, 23, 64)	0	conv1d[0][0]
lstm (LSTM)	(None, 128)	66,560	flow_input[0][0]
flatten (Flatten)	(None, 1472)	0	max_pooling1d[0]…
concatenate (Concatenate)	(None, 1600)	0	lstm[0][0], flatten[0][0]
dense (Dense)	(None, 100)	160,100	concatenate[0][0]
dense_1 (Dense)	(None, 24)	2,424	dense[0][0]

Total params: 230,108 (898.86 KB)

Trainable params: 230,108 (898.86 KB)

Non-trainable params: 0 (0.00 B)

Figure 19: Mixed Neural Network Model for Time series

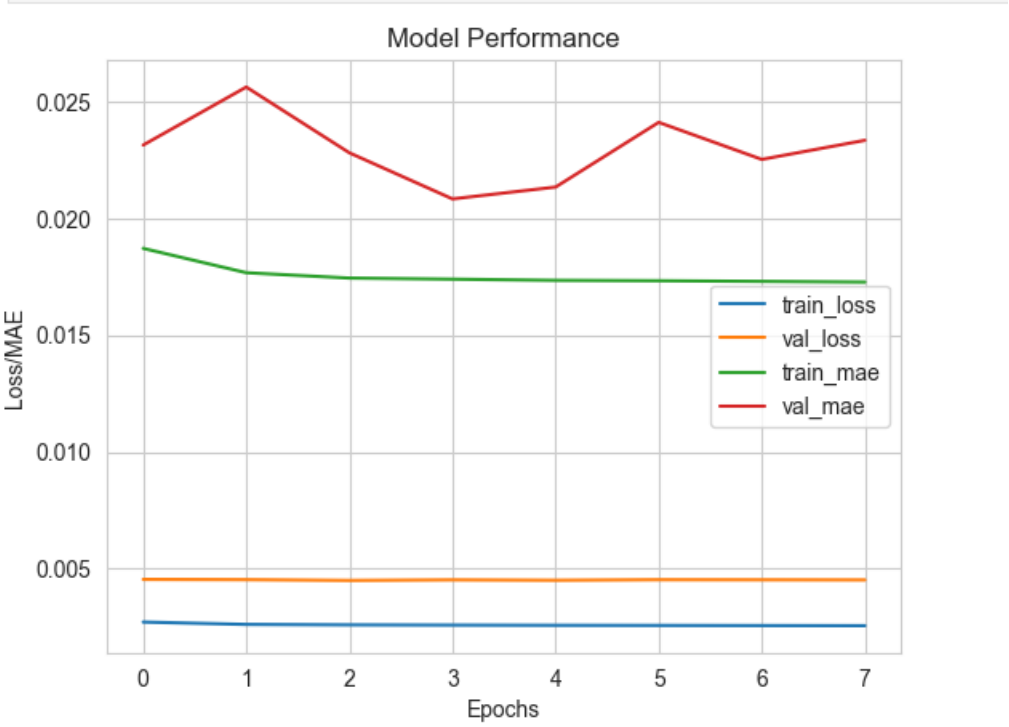

Figure 20: Mixed Neural Network Model Evaluation

- **Validation Loss (val_loss):** Represented in the chart is the loss experienced by the model on the validation set. Initially, there's a slight uptick in the validation loss post the inaugural epoch, after which it stabilizes across subsequent epochs. This pattern suggests that while the model is enhancing its performance on the training data, these improvements are not fully translating to the unseen validation data.
- **Training Mean Absolute Error (train_mae):** This line reflects the Mean Absolute Error (MAE) on the training set, quantifying the average magnitude of errors in the model's predictions. The graph indicates a low MAE level, implying the model's predictions are relatively precise.
- **Validation Mean Absolute Error (val_mae):** This metric, depicted on the graph, shows the MAE on the validation set. It's observed to be higher than the training MAE, which is a common occurrence as models typically exhibit lower performance on unfamiliar data compared to the data they were trained on.

This diagnostic analysis provides crucial insights into the model's learning trajectory and predictive accuracy, highlighting areas of strength and potential improvement.

```
1758/1758 ━━━━━━━━━━━━━━━━━━━━ 11s 6ms/step
MSE: 0.004491092372315573
RMSE: 0.06701561289964879
```

Figure 21: MSE and RMSE

MSE and RMSE Analysis In assessing the model's performance on the validation set, I calculated both the Mean Squared Error (MSE) and the Root Mean Squared Error (RMSE). The MSE, which computes the average of the squared discrepancies between predicted and actual values, serves to accentuate more

substantial errors. For our model, the MSE stands at 0.004491092372315573, denoting a generally minor divergence between the model's predictions and the actual data, which is indicative of reliable accuracy.

The RMSE, derived by taking the square root of the MSE, offers an error metric that aligns with the original data's scale. With an RMSE value of 0.06701561289964879, the model demonstrates commendable predictive precision. However, the true significance of this value becomes clearer when contextualized against the data's scale. Given the diverse range of water usage in this project, the achieved RMSE reflects positively on the model's effectiveness.

Comprehensive Model Evaluation Model Performance Insights

- Low Error Metrics: The model exhibits commendably low MSE and MAE values on the validation set, suggesting a minimal deviation in predictions from actual values. This denotes a high level of predictive accuracy, with errors being kept to a minimum.
- Consistency and Stability: After multiple training iterations, the model demonstrates consistent training and validation loss levels, indicating stable performance across both training and unseen data. This consistency is a hallmark of effective generalization.

Considerations Regarding Overfitting

- Discrepancy in Loss Metrics: A notable gap between training and validation losses, particularly noticeable after several epochs, hints at potential overfitting. This disparity suggests that the model might be overly tailored to the training dataset.
- Loss Trends and Overfitting Indicators: While training loss continues to decline, a plateau in validation loss suggests the model might be learning noise from the training set rather than generalizing the underlying trend, another indicator of overfitting.

Training Process Observations

- Rapid Convergence: The model's swift convergence, evident after just a few epochs, suggests efficient learning, possibly due to an optimally configured optimizer or an apt model architecture.
- Implementation of Early Stopping: The adoption of an early stopping mechanism, which halts training when validation loss ceases to decrease, serves as a safeguard against excessive training and potential overfitting.

6.4.2 Review Process

The model demonstrates commendable learning capabilities, achieving low error rates on the validation set, indicative of accurate predictions. Despite this, there remains a slight risk of overfitting, warranting further optimization efforts. For a comprehensive evaluation of the model's robustness and generalization capacity, testing on a more extensive dataset is advisable. Moreover, the model's suitability should be assessed in the context of specific business needs and the acceptable margin for prediction errors.

6.5 Data Preparation

6.5.1 Data Cleaning

The minutes data is quite large to analyze. Hence, the data is resampled to understand the daily use of water in the house.This is how the daily water consumption in the house looks like:

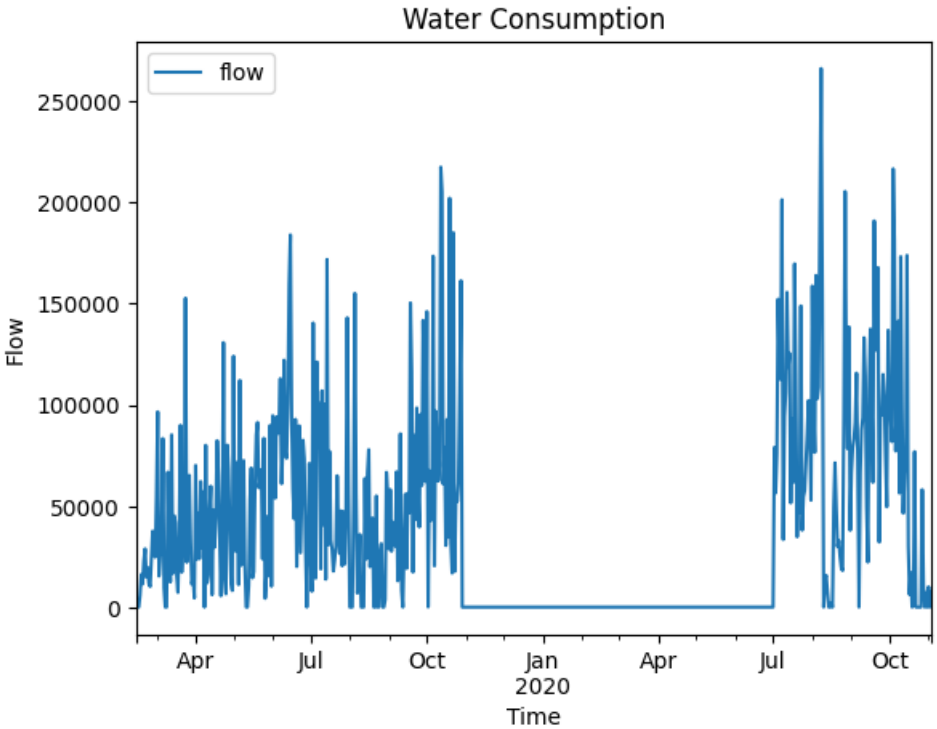

Figure 22: Water consumption in wholehouse

The data goes through the ADFuller test to check whether the time series data is stationary or not. Since the test statistic is more negative than the critical values at all common significance levels (1%, 5%, and 10%) and the p-value is very low, the null hypothesis is rejected of a unit root. This suggests that the time series data is likely stationary.

```
Results of Augmented Dickey-Fuller Test:
(-3.922065546997947, 0.0018783264918908838, 5, 624, {'1%': -3.440873084361836, '5%': -2.866182929627575, '10%': -2.5692425987220
577}, 14531.537889057281)
```

Figure 23: ADFuller test

Next, lags are created in the data to understand the pattern of the flow of utilisation. Lags are created by shifting the original version of values. By reading the past values it becomes easy for the model to understand the dependancies accross time. Now before using the data for modelling, first the data is scaled to noramlise the data at similar lenght. MinMaxScaler uniforms the data in the range of 0 to 1. Next the scaled data is converted to dataframe again for data splitting. After scaling, now the data is splitted into training and testing with 80/20 ratio.

6.6 Modeling

6.6.1 Gradient Boosting Regressor Model Build

For the time series analysis, the Gradient Boosting Regressor Model is used which is based on Gradient Boosting. Gradient Boosting is a powerful boosting algorithm that combines several weak learners into strong learners, in which each new model is trained to minimize the loss function such as mean squared error or cross-entropy of the previous model using gradient descent. In each iteration, the algorithm computes the

gradient of the loss function with respect to the predictions of the current ensemble and then trains a new weak model to minimize this gradient. The predictions of the new model are then added to the ensemble, and the process is repeated until a stopping criterion is met.

```
GradientBoostingRegressor(learning_rate=0.01, max_depth=4, min_samples_split=5,
                          n_estimators=500)
```

Figure 24: Gradient Boosting Regressor Model

The list of parameters used for the model creation is shown above. The model is trained on the features and target of the training data. Next, the model is used to predict against the test data.

6.7 Evaluation

6.7.1 Evaluate Results

```
Gradient Boosting Regressor Mean Squared Error: 0.04192477331004164
Gradient Boosting Regressor Mean Absolute Error:  0.15474302187508146
```

Figure 25: Gradient Boosting Regressor Model MSE and MAE

From above results it can be said that:

Mean Squared Error (MSE): This metric measures the average squared difference between the predicted values by the model and the actual target values. A lower MSE generally indicates a better fit, as it signifies the model's predictions are close to the actual values on average. In this case, an MSE of 0.0419 suggests a good fit.

Mean Absolute Error (MAE): This metric measures the average absolute difference between the predicted values and the actual target values. It provides an idea of the average magnitude of errors made by the model. A lower MAE is desirable, and an MAE of 0.1547 indicates that the model's predictions are on average within 0.1547 units of the true values.

Overall, these metrics suggest that the Gradient Boosting Regressor model is performing well on the regression task. The MSE is low, indicating a close fit between predictions and actual values, and the MAE is also reasonable.

The Below graph shows the predicted values captured by the model against the actual values:

The performance achieved by the model is quite good, but there is one limitation that can be pointed out that there is more deviation in the peak flow values. But other than that, the model performed well in reducing the error rates.

7.Development

In this section we will mention a possible solution that will in return decrease water consumption in general. Our initial thought is to develop a model that will make home owners aware if and only if they are reaching a point of high water consumption. This feature will be made available within a water smart meter were a warning will let the home owner know that the water limit will be exceeded. Hence, by introducing this feature the home owner will see a significant drop of water usage, decrease costs on water bills, and also save water.

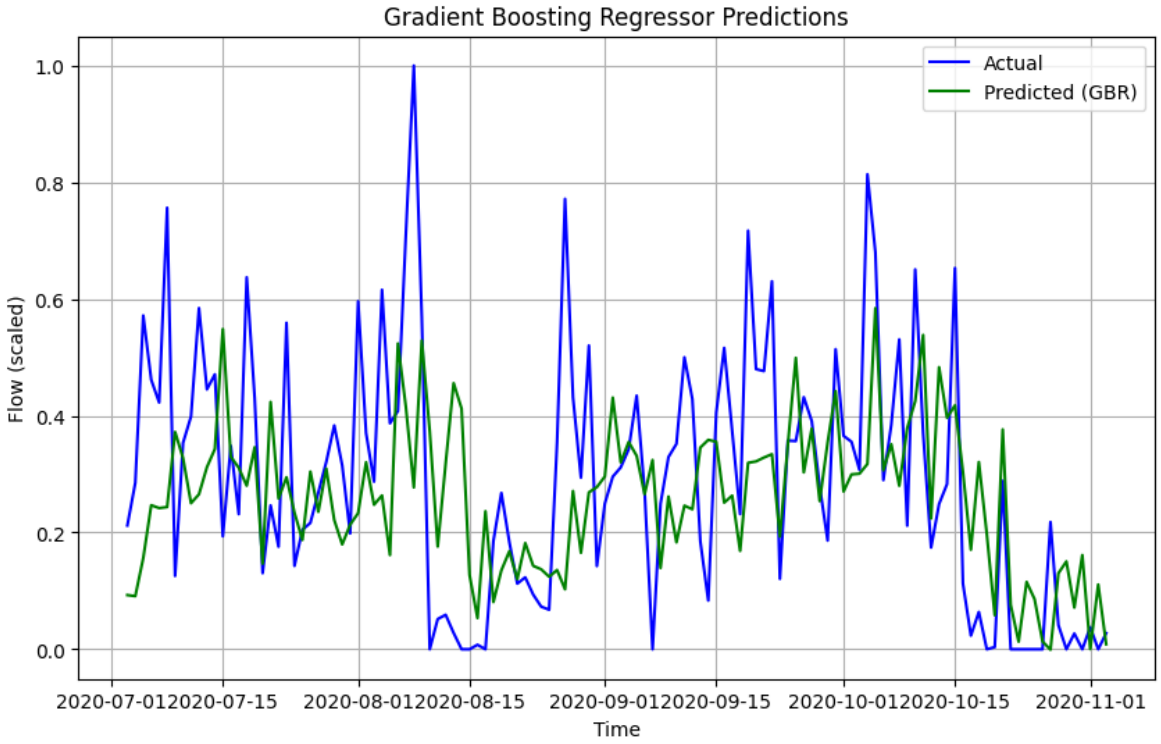

Figure 26: Gradient Boosting Regressor Model Prediction

8.Conclusion

By studying people's usage habits for various appliances, the connection between the equipment used and the water activities during peak and low peak periods, we suggest that water can be used more efficiently and allocated by using water-saving taps, adjusting the pressure or price of water during peak periods, using intelligent water management systems, and cross-designing facilities with high correlated usage.

Reference

[1] Di Mauro A, Di Nardo A, Santonastaso G F, et al. An IoT system for monitoring and data collection of residential water end-use consumption[C]//2019 28th International Conference on Computer Communication and Networks (ICCCN). IEEE, 2019: 1-6.